# Curriculum Vitae

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## Profile

PhD scholar in Natural and Applied Sciences working at the intersection of groundwater systems, hydrogeochemistry, environmental geoinformatics and artificial intelligence. My doctoral work focuses on large-scale prediction and risk assessment of groundwater contamination, integrating groundwater observations with remote sensing, GIS and multi-source environmental data across uranium, fluoride and arsenic contamination. Methods include machine learning, deep learning, automated machine learning and spatial modelling, with broader interests in population exposure, environmental vulnerability, groundwater governance, water security, sustainable development and the cryosphere.

## Education

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| 2024 – present | **PhD, Natural & Applied Sciences**<br>TERI School of Advanced Studies, New Delhi. Coursework CGPA 9.0/10. |
| 2021 – 2023 | **MA Geography**<br>Dr. B.R. Ambedkar University, Agra. CGPA 7.60. |
| 2018 – 2021 | **BA (Honours) Geography**<br>University of Delhi. CGPA 7.86. |
| 2016 – 2018 | **Senior Secondary School**<br>Delhi Public School, Aligarh. CGPA 9.80. |
| 2014 – 2016 | **Secondary School**<br>St. Francis Inter College, Hathras. CGPA 9.20. |

## Doctoral research

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| Thesis | **Development of an Adaptable Machine Learning Based Framework for Multi-Contaminant Prediction** |
| Supervisor | **Prof. Chander Kumar Singh** |
| Status | **Full-time. Registered 1 February 2024. Comprehensive examination completed 28 January 2026.** |
| Coursework | **PhD coursework, CGPA 9.0/10**<br>◦ RPR275 Research and Publication Ethics, 2 credits, grade A<br>◦ RPR271 Research Methodology, 3 credits, grade A<br>◦ NRE115 Environmental Statistics, 4 credits, grade A<br>◦ NRG172 Digital Image Processing & Information Extraction, 4 credits, grade A |
| Framework | **National-scale framework for groundwater contamination prediction and risk assessment**<br>Combines groundwater observations, hydrochemistry, remote sensing, GIS, climate, hydrology, terrain, soil, geology, land use, human modification and population information. Aims: contamination prediction, spatial risk mapping, uncertainty and robustness assessment, environmental controls, exposure assessment and governance decision support. |

## Research experience

2025 – 26

### Uranium in groundwater across India

A national-scale model of where uranium occurs in India's groundwater, built from groundwater observations and environmental predictors and checked against an independent validation dataset. The predictions are turned into spatial risk classes, estimates of population exposure, and governance indicators showing where monitoring and action are most needed.

**Methods:** automated machine learning (AutoGluon), independent validation, spatial risk mapping, exposure assessment.

*Paper submitted to journal; Best Paper Award, ICSSR EHP NEXUS 2026*

2026 –

### Fluoride and the effect of spatial resolution

India-wide prediction of fluoride contamination, benchmarking deep learning models and testing how the resolution of environmental predictors changes the result. Predictor stacks are built in Google Earth Engine at native resolution and at 500 m, 1 km and 5 km, keeping every groundwater observation in the data.

**Methods:** deep learning, multi-resolution predictor extraction, 1.5 mg/L threshold.

*In progress*

2026 –

### Downscaling and upscaling environmental predictors

A physically informed framework for moving satellite data between coarse and fine resolutions without inventing detail. A pilot on CHIRPS rainfall over Punjab (2013–2022) targets 5 km, 1 km and 500 m using sub-pixel fractions and conservation-based aggregation.

**Methods:** machine learning, conservation-based aggregation; validated with Moran's I, semivariograms and aggregation consistency.

*In progress*

2024 – 25

### Arsenic in the Middle Gangetic Plain

The distribution, speciation and controls on arsenic mobilisation in the aquifers of the Middle Gangetic Plain, looking at geogenic and hydrogeochemical factors such as pH, iron, redox potential and sulfate.

**Methods:** hydrogeochemistry, explainable AI (SHAP, LIME).

*Presented at As2024, Bhubaneswar*

2023

### Land surface temperature trends across the Indus River Basin

Spatial and elevation-wise trends in daytime and nighttime land surface temperature across the Indus River Basin.

**Methods:** Google Earth Engine (JavaScript), ArcGIS, Mann–Kendall and Theil–Sen trend analysis.

*Published, Journal of Mountain Science*

2023

### Identification and mapping of permafrost in the Indus Basin

Mapping likely permafrost from about twenty years of mean annual air temperature (MAAT), potential incoming solar radiation (PISR), BMAT, snow and precipitation estimates in Google Earth Engine.

*Paper in progress*

2023

### Snow cover and changes in Indian basins

Changes in snow cover across Indian river basins.

*Paper in progress*

## Publications

2023 **Evaluating spatial and elevation-wise daytime/nighttime LST trends across the Indus River Basin**
Mal, S., **Agrawal, K.**, Rani, S. et al. *Journal of Mountain Science* 20, 3154–3172. doi.org/10.1007/s11629-023-8157-8

2023 **Analyzing the Possibilities and Prospects of 24/7 Water Supply in Delhi, India**
Sanu, S. K., Chandrakanta, **Agrawal, K.**, Sharma, V. R., & Bajaj, A. *Journal of Indian Water Works Association*, LV(1). UGC-CARE Group I, ISSN 0970-275X.

## Manuscripts

2025–26 **Predicting Uranium Occurrences in Groundwater throughout India**
Research paper
*Submitted to journal*

2026 **Comparing and Benchmarking Deep Learning Models for Predicting Groundwater Contamination caused by Fluoride with the effects of Spatial Resolution on Prediction**
Research paper
*In progress*

2026 **A Physically Informed Framework for Multi-Resolution Downscaling and Upscaling of Environmental Predictors (Satellite Data)**
Research paper
*In progress*

## Conference presentations

Apr 2026 **A Machine Learning-Based Framework for Uranium Risk Assessment, Population Exposure and Groundwater Governance in Urban India**
ICSSR EHP NEXUS 2026, Track X: AI and Predictive Analytics for Society and Environment, Gautam Buddha University, 27–28 April 2026. LinkedIn post
*First Position, oral presentation, and Best Paper Award*

Oct 2024 **Distribution, Speciation and Controls on As Mobilization in the Groundwater in Aquifers of the Middle Gangetic Plain, India**
9th International Congress & Exhibition on Arsenic in the Environment (As2024), KIIT Deemed to be University, Bhubaneswar, 20–24 October 2024. Poster.

Nov 2022 **Problems and Prospects of Delhi's Drinking Water Supply**
IGU Thematic Conference on Sustainability, Future Earth and Humanities, Department of Geography, Central University of Haryana, Mahendragarh, 24–25 November 2022.

Jul 2022 **Educational Relevance of Ramayana in Modern India**
National Seminar on Promoting Human Values and Culture for Peaceful Co-Existence, Amity University Haryana, sponsored by ICPR, 22 July 2022.

May 2022 **Digital Enhancement of Physical Space and Connective Turn in Tourism Industry during and Post-COVID-19 Period**
International Conference on Geography and the Digital: Intersections and Conflations, Department of Geography, Delhi School of Economics, University of Delhi, 23–25 May 2022.

May 2022 **Migrant College Students during Covid Pandemic: Evidence from the Field**
Jamia International Conference on Education 2022, Faculty of Education, Jamia Millia Islamia, New Delhi, 6–8 May 2022.

Apr 2022

**Migrant College Students during Covid Pandemic: Evidence from the Field**
International Conference on Challenges to Disaster Risk Reduction and Resilient Habitat, Centre for Disaster Management Studies, Shaheed Bhagat Singh College, University of Delhi, 5–6 April 2022.

## Teaching

Jun–Dec 2026

**Principles of GIS and GNSS (NRG176)**
MSc Geoinformatics, TERI School of Advanced Studies. 4 credits, 60 hours (L-T-P 42-2-32).
*Course teaching*

Jun–Dec 2026

**Principles of Remote Sensing (NRG169)**
MSc Geoinformatics, TERI School of Advanced Studies. 4 credits, 60 hours (L-T-P 42-2-32).
*Course teaching*

Jul–Dec 2025

**Principles of Remote Sensing (NRG169)**
MSc Geoinformatics, TERI School of Advanced Studies. 32 practical learning hours.
*Course taught*

## Projects

### Postgraduate, 2021–2023

- Measuring the Problems and Emotions of Urban-Ward Migrants: A Case Study of Construction Workers in Delhi
- Monitoring the Effects of Light Pollution & Evaluating the Effectiveness of Indian Regulations
- Community Awareness, Participation, & CCDRR: Reflections from North-West India
- Experience of Migrant College Students During COVID-19 Pandemic
- Analyzing the Possibilities and Prospects of 24/7 Water Supply in Delhi, India
- Geography of Ramayana
- Digital Enhancement of Physical Space and Connective Turn in the Tourism Industry during and Post-COVID-19 Period

### Undergraduate, 2018–2021

- Forest Monitoring & Urban Sprawl Analysis Using GIS in Delhi
- LULC Change Detection for 20 Years in South-West Delhi
- Social Indicators & Status Analysis of Jammu & Kashmir
- Socio-Economic Impact of COVID-19 in Delhi
- Snow Cover & Changes in Indian Basins
- HRVC Analysis of Jim Corbett
- Studying Human Trafficking in Delhi through Remote Sensing
- Understanding Ansal Plaza Mall of South Delhi as a Gendered Landscape: A Temporal Study

### Other

- Working With Geospatial Datasets
- High-Quality Research Writing

## Awards and achievements

2026

**First Position, oral presentation, and Best Paper Award**
ICSSR EHP NEXUS 2026, Gautam Buddha University.

2025

**2nd Rank, Map Making Competition**
Amity University.

2018

**State Child Scientist Award**
National Science Congress.

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| 2015 – 2018 | **Consecutive top 3 ranks, English Olympiads**<br>British Olympiad Foundation. |
| 2016, 2018 | **Merit Awards** |
| | **20 cultural and general knowledge awards**<br>From various organisations, including prizes in dance. |

## Rapporteurship and policy engagement

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| Mar 2026 | **Rapporteur, National Workshop on Geospatial Technology for Climate Action, Sustainability and Disaster Risk Reduction**<br>TERI School of Advanced Studies in collaboration with NRSC, ISRO, 16–20 March 2026. |
| Feb 2026 | **Delegate, World Sustainable Development Summit 2026**<br>Taj Palace, New Delhi, 25–27 February 2026. Transformations: Vision, Voices, and Values for Sustainable Development and Climate Action. |
| Mar 2025 | **Rapporteur, Ministerial Session: NDC 3.0 and Multilateralism on the Road to COP30 and COP33**<br>World Sustainable Development Summit 2025, TERI. India Habitat Centre, New Delhi, 6 March 2025. Session summary |
| Mar 2025 | **Rapporteur, High-Level Plenary on Achieving Sustainable Development through Integrated Action**<br>World Sustainable Development Summit 2025, TERI. India Habitat Centre, New Delhi, 7 March 2025. Session summary |

## Positions of responsibility

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| 2026 | **Facilitator and Judge, Programme LEAD 2026**<br>TERI SAS. Sustainable City Design Challenge for students of classes 9–12. |
| Nov 2025 | **Judge and Panel Member, TERI SAS Sustainability Hackathon 2025**<br>Expert evaluation panel for school and undergraduate sustainability proposals, 18–19 November 2025. |
| 2025 | **Judge, KnoxCode 2025, Geospatial Innovation Challenge**<br>Organised by GeoknoX, TERI SAS, as part of Aahwaan Fest 2025. |
| 2018 – 2021 | **Editor-in-Chief, Geography Association**<br>University of Delhi. |
| 2018 – 2020 | **Magazine Coordinator, *Vani***<br>Shaheed Bhagat Singh College, University of Delhi. |
| 2019 | **English Editor, *Vani***<br>Shaheed Bhagat Singh College, University of Delhi. |

## Workshops, seminars and summits attended

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| Feb 2026 | **Water as a Commons workshop, Water Governance Working Group**<br>With the Ostrom Workshop, Indiana University Bloomington and IU India Gateway, 26–27 February 2026; included a Yamuna River field visit with INTACH Delhi. |
| Feb 2026 | **Ambition to Action: Climathon, Delhi Climate Innovation Week**<br>Climate Career & Sustainability Expo, Innovate X Delhi, 21 February 2026. |
| Feb 2026 | **Guest lecture: Soil Carbon Sequestration as a Climate Mitigation Mechanism**<br>Ms. Violeta Martinkė, Project Development Lead, InSoil, 13 February 2026. |

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| Oct 2025 | **Act4Earth Dialogue: Strengthening Multilateralism on the Road to COP30 and Beyond**<br>Virtual, 17 October 2025. |
| Oct 2025 | **Expert talk: DHI Urban Water Solutions**<br>Mr. Morten Just Kjølby, Product Portfolio Manager for MIKE+, DHI Denmark, TERI SAS, 7 October 2025. |
| Sep 2025 | **Science Summit 2025: Marine Resources for Sustainable Development**<br>16 September 2025. |
| Aug 2025 | **QGreenland Educator Workshop**<br>Virtual, 5–7 August 2025. Developing educational activities with QGreenland data. |
| Jul 2025 | **Roots to Resilience Seminar**<br>TERI SAS with UNEP, UNESCO and Akashvani, 25 July 2025. |
| Mar 2025 | **World Sustainable Development Summit 2025**<br>New Delhi, 5–7 March 2025. Partnerships for Accelerating Sustainable Development and Climate Solutions. |
| Oct 2024 | **Lecture by Professor John A. Cherry on groundwater and sustainable development**<br>2020 Stockholm Water Prize laureate, 24 October 2024. Prof. Cherry also spent four hours at the Analytical and Geochemistry Laboratory, TERI SAS, discussing the lab's current and future work. |
| | **Climate Skills: Seeds for Transition**<br>TERI SAS and TERI with British Council and HSBC Bank. |

## Training, capacity building and schools

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| Mar 2026 | **National Workshop on Geospatial Technology for Climate Action, Sustainability and Disaster Risk Reduction**<br>TERI SAS in collaboration with NRSC, ISRO, 16–20 March 2026. Participant, volunteer and rapporteur. |
| Dec 2024 | **Capacity Building Workshop on UAV/Drone Applications in Agriculture and Natural Resource Management**<br>TERI SAS with IIT BHU, IIT Roorkee, Namami Gange, Hokkaido University, GLP Earth, GLP Japan, Abhipsa Foundation and Kambil, 10–12 December 2024. |
| Nov – Dec 2024 | **21-day Winter School Capacity Building Program in Geospatial Science and Technology**<br>Amity University, Noida, supported by the National Geospatial Program, DST, Government of India, 18 November – 7 December 2024. Included a visit to Esri India. |
| Mar 2024 | **Harnessing Geospatial Technology for Forest Fire Disaster Management: Mapping, Monitoring and Mitigation**<br>One-week capacity building programme, Amity University, Noida, 11–15 March 2024. |

## Skills and working knowledge

**Technical skills**

GIS and remote sensing methods and techniques · ArcGIS / ArcGIS Pro · QGIS · Google Earth / Google Earth Engine · ERDAS / ERDAS Imagine · Python · JavaScript for Google Earth Engine · pandas · Machine learning, deep learning and automated machine learning · AutoGluon ·

Random Forest · LightGBM · SHAP and LIME explainable AI ·
Spatial data analysis and visualisation · Time-series modelling and trend analysis ·
Mann–Kendall trend testing and Theil–Sen slope ·
Remote-sensing image enhancement and feature extraction · Orthorectification and mosaicking ·
Satellite-image acquisition, processing and visualisation · DEM processing

**Research and professional skills**

Field-based and survey-based research · Socio-environmental data analysis ·
Groundwater contamination and drinking-water quality assessment ·
Disaster risk reduction and resilience concepts · Climate–water–environment interactions ·
Sustainability and climate policy · Academic writing and conference presentations ·
Scientific communication and reporting · Editorial and content development ·
Rapporteurship and policy-session documentation · Academic coordination and event support ·
Interdisciplinary collaboration

**Remote-sensing and environmental parameters**

MAAT · BMAT · PISR · Forest monitoring · NDVI / NDBI ·
Mann–Kendall trend and Theil–Sen slope · Urban sprawl analysis · Snow estimates · Slope ·
Night-time light · Time-series modelling · MODIS Terra/Aqua · Trend tests ·
Land-use and environmental history · Remote-sensing data repositories ·
Land surface temperature · Image enhancement and feature extraction ·
Orthorectification and mosaicking · Satellite-image acquisition and processing · Precipitation ·
DEM · Landsat · SRTM ·
Climate variables: temperature, precipitation, aerosols, solar radiation and air temperature ·
Cryosphere-related parameters and processes

**Earth observation, environmental and hydrogeochemical data**

Landsat 8/9 · Sentinel · MODIS Terra/Aqua · VIIRS · FIRMS · CHIRPS ·
IMD precipitation products · ERA5-Land · TerraClimate · GLDAS · GRACE · SRTM ·
MERIT Hydro · JRC Global Surface Water · OpenLandMap soil products · GPW · GHM ·
GHSL · SMOD · LCZ · CSP Global Human Modification and related datasets ·
CGWB groundwater and hydrogeochemical observations ·
Literature-derived groundwater quality observations ·
Hydrochemical variables: pH, EC, Ca, Mg, Na, K, $HCO_3$, Cl, $SO_4$, $NO_3$, fluoride, phosphate and uranium

**Data management, QA/QC and research reliability**

Duplicate detection and duplicate-preservation rules ·
Administrative-name normalisation and standardisation ·
Coordinate validation and geometry reconstruction · Sentinel-value handling ·
Missing-value identification · Source preservation and data lineage/provenance ·
Unit verification · Temporal and spatial compatibility checks · Resolution harmonisation ·
Independent validation · Masked-pixel handling ·
Denominator consistency in exposure estimates · Correlation versus causation in interpretation ·
Caution about Pearson correlation as a hard feature filter ·

Ground truth, transferability, uncertainty and physical plausibility ·
Temporal mismatch between monitoring observations and environmental predictors ·
Resampling methodology: true downscaling versus interpolation ·
Geogenic versus environmental controls in groundwater contamination

## Subjects studied

Geomorphology · Climatology & Oceanography · Cartographic Techniques ·
Thematic Cartography · Remote Sensing & GIS · Statistical Methods in Geography ·
Field Work & Research Methodology · Academic Writing & Composition · Geography of India ·
Human Geography · Disaster Management · Media & Communication Skills ·
History of Geographical Thought · Social Geography · Political Geography ·
Economic Geography · Agricultural Geography · Population & Settlement Geography ·
Environmental Geography · Regional Planning & Development

## Future research interests

- **Permafrost**: identification, modelling and assessment of permafrost.
- **Glaciers, ice and snow**: mapping and calculations of glacier, ice and snow change.
- **Climate change**: assessments of climate change, its impacts and its parameters.
- **High Mountain Asia and the poles**: the Himalaya, High Mountain Asia, the Arctic and Antarctica.
- **History and archaeology**: remote sensing assessments and calculations for history and archaeology.

## Volunteering and co-curricular

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| 2016 – present | **Volunteer, SWAPO NGO** |
| 2019 – 2021 | **National Service Scheme (NSS)** |
| 2018 – 2019 | **Coordination of Volunteers, Vasundhara** |
| 2018 – 2023 | **Personal blogging and content writing** |

## Personal interests

Collecting old coins and stamps, reading historic stories and accounts, photography and videography, travelling, badminton, dance, debating and Model UN.

## Research keywords

Groundwater contamination; groundwater quality; groundwater quantity; hydrogeochemistry; uranium; fluoride; arsenic; groundwater risk assessment; population exposure; environmental vulnerability; groundwater governance; water security; environmental geoinformatics; geospatial data science; remote sensing; GIS; Google Earth Engine; machine learning; deep learning; AutoML; explainable AI; spatial modelling; spatial statistics; multi-resolution modelling; downscaling; upscaling; Earth observation; hydrology; climate–water interactions; hydro-environmental systems; cryosphere; permafrost; snow; glacier-fed hydrology; LST; climate change; sustainability; disaster risk reduction; policy engagement; environmental inequality; exposure assessment.

## Contact

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